Asset Performance:

Pol 04 Global warming potential of refrigerants

4 Credits

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No Minimum Standard

Aim

To encourage the use of refrigerants with a low global warming potential (GWP) in refrigerant equipment.

Question

What refrigerants are used in the asset refrigeration equipment?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	Some refrigerants have a global warming potential of >10 (e.g. majority HFCs, HCFC, CFCs)
1	C.	50%, by kW cooling or heating capacity, of refrigerants have a global warming potential of ≤10 (e.g. Propane, Butane)
2	D.	All refrigerants have a global warming potential of ≤10 (e.g. Propane, Butane)
4	E.	All refrigerants have a global warming potential of ≤1 (e.g. Ammonia, Water, Carbon dioxide)

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	**Filtering** Where no refrigerants are used or only small hermetic systems (refrigerant charge in each system is ≤5kg) are installed in the asset, this issue can be filtered out of the assessment.	All
2.	The issue applies to all equipment and areas using refrigerants including, but not limited to: a) Walk-in cold storage enclosures b) Cold storage, including commercial food/drink display cabinets but excluding residential-scale white goods (e.g. fridges and freezers) c) Comfort cooling and heating (e.g. heat pumps)	All

Criterion	Assessment criteria	Applicable Answer
	d) Process based cooling loads (e.g. servers/IT equipment)	
3.	A list of typical refrigerants with a low GWP can be found in Table 27 in the Checklist and Tables section of this issue.	All
4.	This issue applies to refrigerants used in equipment that is installed on-site only.	All

Checklists and Tables

Table 27: Common refrigerant types with a low GWP

R-Number	Chemical name	GWP 100-yr
R-30	Dichloromethane	9
R-170	Ethane	3
R-290	Propane	3
R-600	Butane	3
R-600a	Isobutane	3
R-717	Ammonia	0
R-718	Water	<1
R-729	Air (nitrogen, oxygen, argon)	0
R-744	Carbon dioxide	1
R-1150	Ethylene	3
R-1234yf	2,3,3,3-Tetrafluoropropene	<1
R-1270	Propylene	3
Sources: The United Nations Environment Programme (UNEP) '2010 Report of the Refrigeration, Air-conditioning and Heat Pumps Technical Options Committee' EN 378-1:2016 Refrigerating systems and heat pumps. Safety and environmental requirements. Basic requirements, definitions, classification and selection criteria The Intergovernmental Panel on Climate Change 5th Assessment Report, Chapter 8, 'Anthropogenic and Natural Radiative Forcing', 2013		

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Copy of manufacturer's information confirming the global warming potential of refrigerants used on-site.
All	Photographic evidence of refrigerant packaging/systems (if necessary).
1	Statement from the building manager indicating that the asset does not contain any systems that contain refrigerants or confirmation that the total charge is ≤5kg in any systems that are present.

Definitions

GWP:

GWP is defined as the potential for global warming that a chemical has relative to 1 unit of carbon dioxide, the primary greenhouse gas. In determining the GWP of the refrigerant, the Intergovernmental Panel on Climate Change (IPCC) methodology using a 100-year Integrated Time Horizon (or ITH) should be applied.

Refrigerant:

A compound typically found in either a fluid or gaseous state that readily absorbs heat from the environment when combined with other components such as compressors and evaporators

Additional information

Refrigerants

There are three main make-ups of refrigerants:

a) **Hydrogenated Fluorocarbon Refrigerants (HFCs)** are made up of hydrogen, fluorine, and carbon. Because they do not use a chlorine atom (which is used in most refrigerants) they are known to be one of the least damaging to our ozone.

b) **Hydrogenated Chlorofluorocarbon Refrigerants (HCFCs)** are made up of hydrogen, chlorine, fluorine, and carbon. These refrigerants contain minimal amounts of chlorine; they are not as detrimental to the environment as some other refrigerants.

c) **Chlorofluorocarbon Refrigerants (CFCs)** contain chlorine, fluorine and carbon. These refrigerants carry high amounts of chlorine, so they are known for being the most hazardous to the ozone layer.

Hydrocarbons and ammonia-based refrigerants have low or zero GWP. These are now widely available and are valid alternatives to HFCs in all buildings, providing health and safety issues are fully addressed. United Nations Environment Programme (UNEP) hosts a HCFC Help Centre which contains information about the management and phase out of HCFCs and alternatives to HCFCs. Table 27 contains a list of common refrigerant types with a low GWP.